

- We have multiple end-points. So need to make different **response** and **request** DTOs.
 - ⌘ Previously we were using same DTO for request and response. But now we'll not.

```
dto.auth
  AuthResponse
  LoginRequest
  SignupRequest
  UserProfileResponse
```

- ⌘ **AuthResponse** = **UserProfileResponse** + **token**.

```
AuthResponse(String token, UserProfileResponse user) {
```

- ⌘ **LoginRequest** and **SignupRequest** are DTOs for POST request payload.
- ⌘ You should create different DTOs for request and response.
- ⌘ For **getProfile** controller, only **UserProfileResponse** is needed, there is no need of token.

- We'll use **MapStruct** for mapping of one class type to another instead of **ModalMapper**.

- ⌘ As we are using **lombok** as well, so we need to make sure that **lombok** will be executed first then **MapStruct**.
- ⌘ Because, both will be executed during *compile time*.
- ⌘ Annotation processors do NOT run sequentially like "Lombok first, MapStruct second". They run in the same compilation round.
- ⌘ For this **lombok-mapstruct-binding** is required.

```
<properties>
  <java.version>21</java.version>
  <org.mapstruct.version>1.6.3</org.mapstruct.version>
</properties>
```

(variable)

```
<dependency>
  <groupId>org.mapstruct</groupId>
  <artifactId>mapstruct</artifactId>
  <version>${org.mapstruct.version}</version>
</dependency>
<dependency>
  <groupId>org.projectlombok</groupId>
  <artifactId>lombok-mapstruct-binding</artifactId>
  <version>0.2.0</version>
</dependency>
</dependencies>
```

➤ **Dependency vs Plugins**

- ⌘ Dependencies are used by your application at compile/runtime.
- ⌘ Plugins are used by Maven to build your application.

➤ To create mapper (MapStruct),

```
@Mapper(componentModel = "spring")  
public interface ProjectMapper {  
}
```

- ⌘
- ⌘ Use the **componentModel="spring"**